



Topic: Conceptual Modeling

A Case Study

TechMarket, SA sells electronic equipment online on a dropshipping basis, not keeping its own stock. When a customer places an order, the company registers the order and forwards it to the supplier by email, who ships the product directly to the customer. Despite the diversity of brands it represents, it is computers and smartphones that guarantee the company most of its turnover. Computers can be laptop or desktop, with different processor and storage configurations. Smartphones, on the other hand, are distinguished by the type of operating system (android/ios), battery capacity and memory.
(...)

TechMarket, SA									
Address: Rua da Tecnologia					Client: 123456				
Phone.: 912345678					Name: Cristiana Neto				
NIF: 123456789					NIF: 234567890				
					Phone: 923456789				
					Address: Rua Aleatória, 33, 4810-123				
Order N.º:123456			Order Date: 23/02/2026						
N.º	Code	Type	Description	Fornecedor	Preço Unitário	Qtd	Desconto	Preço	Observações
1	C12	Laptop	Laptop Lenovo i5 16GB/512GB Grey	ElecSupply Ltd	899,00 €	1	10,00 €	889,00 €	Scratch in the box
2	T14	Smartphone	iPhone 15 Pro 256GB Black	MobileTraders	1 119,00 €	1	- €	1 119,00 €	Includes protective film
3	C5	Laptop	Macbook Pro 16GB/512GB Black	ElecSupply Ltd	2 000,00 €	2	- €	4 000,00 €	
						4	10,00 €	6 008,00 €	

Figure 1 - Example of a purchase order

Based on the case study presented, it is intended to develop a conceptual model for a relational database, capable of representing the information associated with the orders (Figure 1) described in the statement. In addition to the model, it must indicate the requirements considered in its construction, as well as present the characterization tables that make up the respective data dictionary. The resolution must necessarily follow the different stages of the methodology for the development of conceptual schemes addressed in the course.